Priyanka Tiwari

+91 9019938267 priyankatiwari140419@gmail.com linkedin.com/in/priyanka-tiwari-2b0992263 github.com/Princess0407

PROFESSIONAL SUMMARY

Fourth Year ECE student with strong Python and embedded systems expertise. Shipped provtrack to PyPI with 94% test coverage; designed ICELang compiler with full AST pipeline; built GXU-1 GPU co-processor. Proficient in object-oriented design, automated testing, and version control. Seeking renewable energy internship to apply software testing, MATLAB, and control systems validation expertise.

CORE COMPETENCIES

Languages: C/C++, Python, JavaScript, TypeScript, SQL, MATLAB, HTML/CSS
ML / AI: TensorFlow Lite Micro, on-device inference, model quantization (INT8), edge AI, split inference, NumPy, pandas, data analysis
Compiler / Systems: DSL design, AST construction, lexer/parser pipelines, PLY (Lex-Yacc), interpreter design
Embedded & Hardware: ESP32/ESP32-S3, FreeRTOS, R-2R DAC, VGA signal generation, KiCad, Ngspice, TinyML
Frameworks & Tools: React, Vite, Next.js, Flask, FastAPI, Supabase, Matplotlib, Docker, WebSocket, REST APIs, CI/CD pipelines
Developer Tools: Git, Linux, VS Code, KiCad Scripting API, WebGL, technical documentation

PROFESSIONAL EXPERIENCE

FOSSEE Summer Intern | May 2026 – Present
IIT Bombay – Free and Open Source Software for Education *Remote*

- Designed and implemented ICELang, a DSL for hardware description, with a complete lexer/parser pipeline and AST, reducing manual schematic authoring time for structured hardware definitions.
- Accomplished end-to-end KiCad schematic export by translating ICELang AST nodes into native `.kicad_sch` format, automating symbol placement, wire routing, and pin mapping at ∼90% layout accuracy.
- Developed an SVG symbol rendering engine for pre-export visual verification of schematic components, eliminating repeated KiCad round-trips and measurably reducing debug cycle time.
- Integrated 50+ standard component symbols and authored test suites achieving >85% coverage, surfacing 12+ edge-case bugs before milestone submission.
- **Stack:** Python, PLY (Python Lex-Yacc), KiCad Scripting API, SVG rendering, AST design patterns, Git

PROJECTS

provtrack: Zero-Instrumentation ML Pipeline Provenance Tracker | 2026
Python, NetworkX, SHA-256, pandas, scikit-learn, PyPI

- Built a 4-layer runtime proxy (proxy → hasher → logger → graph) that instruments an entire ML session with one `activate()` call – zero decorators, zero config files, zero pipeline rewrites required.
- Engineered SHA-256 fingerprinting of every intermediate DataFrame state into a cryptographically-linked DAG; reservoir sampling caps hash latency at ∼25ms on 1M+ row inputs with deterministic cross-run hashes.
- Achieved 94% test coverage across 64 tests and 3 suites, documenting 15 edge cases (pickle bypass, inplace mutation, multi-threaded logger safety); shipped `report / export / query / diff / validate` CLI.
- Published to PyPI (`pip install provtrack`) under Apache 2.0; immutable frozen dataclass records and fully reversible `deactivate()` for production-safe deployment with zero use of `eval`/`exec`.

GXU-1: Custom GPU Co-Processor | 2026
ESP32-S3, FreeRTOS, C, React, Vite, WebSocket

- Architected a custom GPU co-processor on the ESP32-S3 with a handbuilt R-2R resistor ladder DAC for analog VGA signal output, achieving functional display output under a Rs.700 bill of materials.
- Designed a custom 8-opcode binary ISA (DRAW, FILL, CLEAR, BLIT, RECT, LINE, PIXEL, NOP) interpreted by a software rasterizer, achieving ∼30 FPS via dual-core FreeRTOS (Core 1: render, Core 0: host comms).
- Built a React/Vite WebSocket dashboard with live VRAM heatmap telemetry, opcode throughput graphs, and a browser-based ISA emulator enabling hardware-free development.

OPEN SOURCE CONTRIBUTIONS

Laya – AI Notification Command Center (Tauri + Python) | PR #40 2026

- Closed unauthenticated loopback access to Laya's engine REST API by designing OS keychain-backed bearer token authentication across 3 new security modules (`keychain.py`, `engine_auth.py`, updated `main.py`), covering all Tauri shell and Svelte UI requests.
- Introduced a dedicated `laya_engine_api_token` keychain entry distinct from the existing `laya_mcp_bearer`, preventing token reuse across trust boundaries; all protected endpoints return `HTTP 401` on missing or incorrect tokens.

ACHIEVEMENTS

Microsoft Codess Mentee 2025 – Selected for Microsoft's mentorship program for women in technology; chosen for technical merit and leadership potential.

HPAIR 2026 Delegate – Selected as a delegate for the Harvard Project for Asian and International Relations; placed in the top 1% of the global applicant pool.

EDUCATION

Acharya Institute of Technology Bangalore, IN
Bachelor of Engineering in Electronics and Communications Engineering Aug 2023 – May 2027